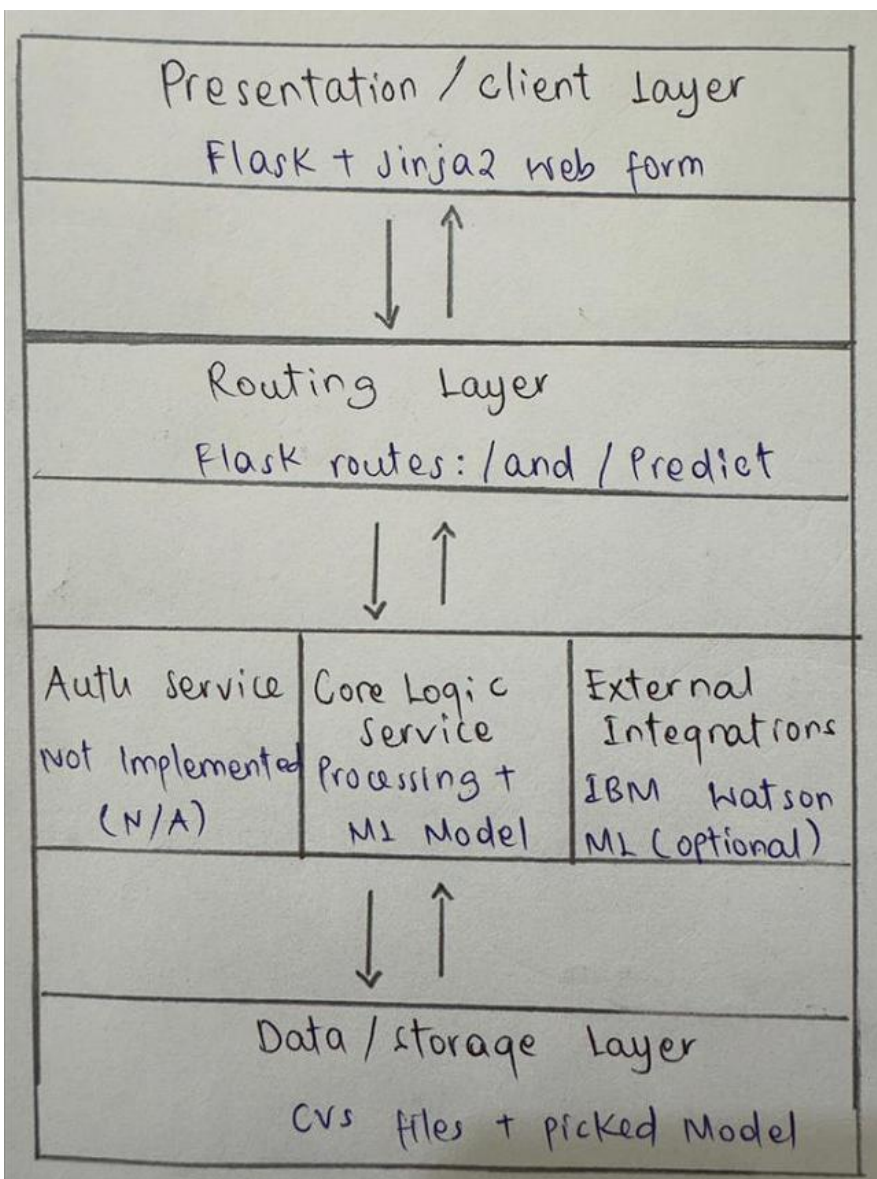


Solution Architecture

Date	5th July 2026
Team ID	RAHMA HUSSEIN JUMA
Project Name	CREDIT CARD APPROVAL PREDICTION
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects applicant details through a web form and displays the approval/rejection result to the user	HTML, CSS, Jinja2 (Flask templates)

Routing Layer	Handles incoming requests and directs them to the right function — no separate gateway/load balancer needed at this scale	Flask (built-in routing)
Core Logic Service	Applies the same preprocessing used in training, then runs the trained model to predict approve/reject	<i>Python, scikit-learn, XGBoost, pandas</i>
External Integrations	Optional cloud hosting of the trained model for scalable, real-time predictions	IBM Watson Machine Learning
Database / Storage	Stores the source datasets, the processed training data, and the saved model artifacts (model, scaler, column list)	CSV files, Pickle (.pkl)